Program 4 — Complete Power BI Practical Guide From Scratch

Program title: Enhancing Customer Experience with Predictive Analytics and Data Mining by Taking a Suitable Business Scenario (Microsoft Power BI).
You are using seven CSV files:

File	Purpose
sales.csv	Sales transactions, quantity, price and sales details
products.csv	Product information
customers.csv	Customer information
categories.csv	Product categories
employees.csv	Employee information
cities.csv	City information
countries.csv	Country information

Phase 1 — Data Ingestion and Cleaning
Step 1: Open Power BI and load the files

1. Open Power BI Desktop.
2. Click Home → Get data → Text/CSV.
3. Select sales.csv.
4. Click Open.
5. In the preview window, click Transform Data.
6. Repeat for the other six CSV files. If Power BI opens Power Query, use Home → New Source → Text/CSV to add the remaining files.

Step 2: Rename tables
In Power Query Editor, use the Queries pane on the left.
Rename the queries to:

- Sales
- Products
- Customers
- Categories
- Employees
- Cities
- Countries

Right-click a query name and select Rename.

Step 3: Clean the data
For each table:

1. Select the table.
2. Select the relevant columns and use Home → Remove Rows → Remove Duplicates only when removing duplicate records is appropriate.
3. Rename columns if necessary to remove spaces or make names consistent.
4. Check the data types of the columns.

For the Sales table:

- quantity → Whole number, if quantities are whole numbers.
- unit_price → Decimal number or Fixed decimal number.
- sales_date → Date, if present.
- Replace null values in quantity or unit_price with zero only if that is appropriate for the source data.

Your earlier doubt: Do not create a missing column just because the PDF mentions it. You confirmed that quantity and unit_price are available in your Sales table, so use those actual columns.

Step 4: Add date fields
If sales_date exists and is a valid Date column:

1. Select Sales in Power Query.
2. Select sales_date.
3. Click Add Column → Date.
4. Add Month, Quarter and Year if required.

Click Home → Close & Apply.
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Phase 2 — Data Modeling

Step 1: Open Model view
Click the connected-tables icon on the left side of Power BI.

Step 2: Check these six relationships

Table A	Column	Table B	Column
Sales	customer_id	Customers	customer_id
Sales	product_id	Products	product_id
Products	category_id	Categories	category_id
Sales	employee_id	Employees	employee_id
Customers	city_id	Cities	city_id
Cities	country_id	Countries	country_id

Save your project using Ctrl + S.

Phase 3 — Customer Segmentation
Follow the corrected method below.
Step 1: Create Transaction Amount
You confirmed that Sales contains quantity and unit_price.

1. Open Data view or Model view.
2. Right-click Sales in the Data pane.
3. Click New column.
4. Enter:

Transaction Amount =
Sales[quantity] * Sales[unit_price]
5. Press Enter.
If this column already exists and works correctly, do not create another one.

Step 2: Create Total Spend Measure
In your earlier attempt, creating Total Spend as a calculated column caused a circular dependency. Use a measure instead.

1. Click Home → New measure.
2. Enter:

```
Total Spend Measure =
SUMX(
   Sales,
   Sales[quantity] * Sales[unit_price]
)
```

3. Press Enter.
Important: You already created this measure. If it exists and works, reuse it. Do not create another measure with the same name.
This measure calculates total revenue in the current filter context. When customer ID is included in a table, the existing relationship should allow it to calculate spending per customer.

Step 3: Understand customer segmentation(optional)
If you need the exact spending-based donut chart, use Modeling → New table and create this separate table:

```
Customer Spend Summary =
ADDCOLUMNS(
   VALUES(Customers[customer_id]),
   "Total Spend",
      CALCULATE(
         SUMX(
            Sales,
            Sales[quantity] * Sales[unit_price]
         )
      ),
   "Spend Segment",
      VAR SpendAmount =
         CALCULATE(
            SUMX(
               Sales,
               Sales[quantity] * Sales[unit_price]
            )
         )
      RETURN
         SWITCH(
            TRUE(),
            SpendAmount > 10000, "High Value",
            SpendAmount >= 5000, "Medium Value",
            "Low Value"
         )
)
```

This is a separate calculated table, not a new column in Customers. It should not reference your earlier Total Spend column. If Power BI reports an error because your model has different field names or relationships, stop and check those names.

Step 4: Create the customer segmentation donut chart

1. Open Report view.
2. Click a blank area.
3. Select Donut chart.
4. Drag Spend Segment from Customer Spend Summary into Legend.
5. Drag customer_id from Customer Spend Summary into Values.
6. Set the aggregation to Count if needed; each row represents one customer.

This chart shows the distribution of High Value, Medium Value and Low Value customers, provided the spend summary calculates correctly.

Step 5: Create a customer count measure
Click Home → New measure and enter:

Customer Count KPI =
DISTINCTCOUNT(Customers[customer_id])
You already created a measure with this name. Reuse it if it exists.

Step 6: Create customers by country
1. Open Report view.
2. Insert a Clustered bar chart.
3. Add Countries[country] to the Y-axis.
4. Add Customer Count KPI to the X-axis.

Step 7: Create the Top 5 Customers table
1. Insert a Table visual.
2. Add Customers[customer_id].
3. Add Total Spend Measure.
4. In the Filters pane, find customer_id under Filters on this visual.
5. Change the filter type to Top N.
6. Enter 5.
7. Drag Total Spend Measure into By value.
8. Click Apply filter.
9. Sort Total Spend Measure in descending order.

This shows the five highest-spending customers, provided the customer relationships correctly filter Sales.

Phase 4 — Product and Category Insights
Objective: Compare products and categories using a Matrix, a Top 10 chart, a slicer, and conditional formatting.
You have already created the Matrix visual. Reuse it if it works.

Step 1: Create the Matrix
1. Open a new report page and rename it Product Insights if you have not already done so.
2. Select the Matrix visual.

Add these fields:

Field	Matrix section
Categories[category_name]	Rows
Products[product_name]	Columns
Sales[total_sales]	Values

If total_sales is not available or does not represent revenue, use the appropriate revenue measure instead.

Step 2: Create the Top 10 Products chart
1. Click a blank area.
2. Select Clustered bar chart.
3. Add Products[product_name] to the Y-axis.
4. Add Sales[quantity] to the X-axis.
5. Ensure the aggregation is Sum.
6. In the Filters pane, expand product_name under Filters on this visual.
7. Change Filter type to Top N.

8. Enter 10.
9. Drag quantity into By value.
10. Click Apply filter.
11. Sort by Sum of quantity in descending order.

Step 3: Add the Category slicer

1. Click a blank area.
2. Select Slicer.
3. Drag Categories[category_name] into the slicer.
4. Place it at the top of the page.

Test it by selecting a category. The connected visuals should filter accordingly.

Step 4: Conditional formatting for quantity below 100

You successfully reached the Background color dialog earlier, but the selected field was Sum of total_sales. That is revenue, not quantity.

To follow the PDF exactly:

1. Select the Matrix visual.
2. Add Sales[quantity] to Values if it is not already present.
3. In the Values field well, open the dropdown beside Sum of quantity.
4. Choose Conditional formatting → Background color.
5. Set Format style to Rules.
6. Base the rule on Sum of quantity.
7. Set the rule to highlight values less than 100.
8. Choose a highlight color and click OK.

If your version displays the formatting options under Format visual → Cell elements, use that route instead.

Save using Ctrl + S.

Phase 5 — Employee Sales Analytics

Objective: Compare employees based on revenue and unique customers served.

You mentioned that your Employees table does not have an employee name column. Use employee_id instead; do not invent employee names.

Step 1: Create the employee table

1. Add a new report page named Employee Performance.
2. Insert a Table visual.
3. Add Employees[employee_id].

Create these measures using Home → New measure. Use the exact names below only if they do not already exist.

Total Revenue KPI

You already created this measure:

```
Total Revenue KPI =
SUMX(
    Sales,
    Sales[quantity] * Sales[unit_price]
)
```

Reuse it rather than creating it again.

Orders Count KPI

Your PDF refers to Order ID. If your Sales table has sales_id but no order_id, use the actual available ID field:

Orders Count KPI =
DISTINCTCOUNT(Sales[sales_id])
This counts distinct sales IDs, which may not be the same as order count if multiple sales rows belong to one order.

Unique Customers Served KPI

Unique Customers Served KPI =
DISTINCTCOUNT(Sales[customer_id])
Add these measures to the table along with employee_id.

Step 2: Create the employee comparison chart

1. Insert a Clustered bar chart.
2. Add Sales[employee_id] to the Y-axis.
3. Add Total Revenue KPI to the X-axis.

Use the employee ID field from the Sales table if the Employees table does not filter Sales correctly.

Step 3: Create the employee KPI cards

You already created measures for:

- Highest Revenue Employee
- Highest Customer Reach Employee

Add a Card visual for each and drag the corresponding measure into it.

Phase 6 — Forecasting Sales

Objective: Show historical sales trends and forecast the next three months.
You have already reached the Forecast option in the Analytics pane.

Step 1: Create the line chart

1. Add a report page named Forecast & KPIs, if it does not already exist.
2. Insert a Line chart.
3. Add Sales[sales_date] to the X-axis.
4. Add Sales[total_sales] to the Y-axis.
5. Ensure total_sales is summarized by Sum.

If the date column is missing or is not typed as Date, fix it in Data view or Power Query first.

Step 2: Find the Analytics pane

You had difficulty locating Analytics. On your version of Power BI:

1. Select the line chart on the canvas.
2. Look at the Visualizations pane.
3. Click the Analytics icon (magnifying glass over a chart), if available.

If you see the Forecast section, expand it.

Step 3: Configure Forecast

Set the options as follows:

Option	Setting
Forecast length	3
Units	Months

Confidence interval	95%
Seasonality	Auto
Ignore last	0

Click Apply.
The chart should show a forecast extending beyond the historical sales line, with a shaded confidence interval.

If Forecast is missing

- Make sure the selected visual is a Line chart.
- Use a valid date column on the X-axis.
- Set the X-axis type to Continuous, if available.
- Check whether the date hierarchy is being used instead of the actual date field.

If the Forecast option still does not appear, do not waste time repeatedly clicking the same icon. Your Power BI version or visual configuration may not support that option in its current state.

Step 4: Add Smart Narrative

1. Click a blank area.
2. Select Insert → Smart narrative, if available.
3. Allow Power BI to generate a text summary.
4. Check the summary against the actual chart.

Save the project.

Phase 7 — Business KPIs

Your PDF specifies Total Revenue, Average Order Value, Customer Count, Returning Customer Percentage and Sales by Region.
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You have already created several measures. Reuse the working ones instead of duplicating them.

Step 1: Total Revenue

Use the existing Total Revenue KPI measure.

Step 2: Average Order Value

You created this measure earlier:

```
Average Order Value KPI =
DIVIDE(
    [Total Revenue KPI],
    DISTINCTCOUNT(Sales[sales_id]),
    0
)
```

This is average revenue per distinct sales ID, assuming sales_id represents the intended order identifier.

Step 3: Customer Count

Reuse Customer Count KPI if it exists.

Step 4: Returning Customer Percentage

You created this measure earlier:

```
Returning Customer Percentage KPI =
VAR RepeatCustomers =
    COUNTROWS(
```

```
        FILTER(
            VALUES(Sales[customer_id]),
            CALCULATE(
                DISTINCTCOUNT(Sales[sales_id])
            ) > 1
        )
    )
VAR AllCustomers =
    DISTINCTCOUNT(Sales[customer_id])
RETURN
    DIVIDE(
        RepeatCustomers,
        AllCustomers,
        0
    )
```

Select the measure and set its format to Percentage using the Measure tools ribbon.
This counts customers with more than one distinct sales ID, assuming each sales ID represents a transaction.

Step 5: Display the KPI cards

Create or reuse Card visuals for:

- Total Revenue KPI
- Average Order Value KPI
- Customer Count KPI
- Returning Customer Percentage KPI

If you see the error “Choose a different name”, it means that measure name already exists. Either select the existing measure or give a new measure a unique name.

Step 6: Create a regional revenue donut chart

1. Insert a Donut chart.
2. Add the available country or region field to Legend.
3. Add Total Revenue KPI to Values.

Your dataset may have a country field but no separate region field. In that case, group by country and describe it as revenue by country, rather than by region.

Phase 8 — Final Dashboard and Storytelling

Page	Main contents
Customer Analytics	Customer segments, customers by country, Top 5 customers
Product Insights	Matrix, Top 10 products, category slicer, conditional formatting
Employee Performance	Employee table, revenue comparison, employee KPI cards
Forecast & KPIs	Sales line chart, forecast, Smart Narrative, business KPI cards

Step 1: Add slicers

Add these slicers to the relevant pages:

Slicer	Field
Country	Countries[country]
Category	Categories[category_name]
Customer Segment	Customers[segment]

Remember that Customers[segment] is your existing category field. It is not automatically the same as the spending-based High Value, Medium Value and Low Value classification.

Step 2: Add a text box

1. Click Insert → Text box.
2. Enter a short explanation of your report.

For example:
“ This report analyses customer spending, product performance, employee sales and sales trends to support business decision-making.”
3. Position the text box neatly at the top of the appropriate report page.

Step 3: Test the report

- Select a country and check whether the connected visuals respond.
- Select a product category and check the Matrix and Top 10 chart.
- Check that the KPI cards display numeric values.
- Check that the sales line chart displays dates correctly.
- Verify that the Top 5 and Top 10 filters are applied.
- Confirm that there are no error messages or circular dependencies.

Step 4: Save the final project

Press Ctrl + S.
Save your file as:
Program4_Customer_Analytics.pbix

Common doubts — quick answers

1. Relationship lines already exist. Should I create them again?
No. Verify the existing relationships and continue if they are correct.
2. Why did Circular dependency appear?
Power BI detected a calculation dependency loop involving the calculated customer-spending or segment columns. Use the separate Customer Spend Summary table described in Phase 3 rather than repeatedly recreating those columns.
3. Why can't I find Distinct count?
If the dropdown is difficult to locate, create a measure with DISTINCTCOUNT() and use that measure in the chart.
4. Why can't I find the Matrix visual?
The Matrix icon looks like a grid with highlighted cells, unlike the plain Table icon.
5. Why can't I find Conditional formatting?
Select the Matrix, add the numeric field to Values, and open the dropdown beside that field. You can also check Format visual → Cell elements if available.
6. Why can't I find the Analytics pane?
Select the Line chart first. Look for the magnifying-glass-over-chart icon in the Visualizations pane. Forecast availability depends on the visual and axis configuration.
7. Why does Power BI say “Choose a different name”?
A measure or column with that name already exists. Reuse it or choose a unique name, such as Customer Count KPI.
8. What if Employee Name is missing?
Use employee_id. Do not invent names that are not present in the dataset.

Your final lab checklist

Program 4 completion
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Phase 1: Import and clean seven CSV files
Phase 2: Verify six relationships
Phase 3: Customer spending and segmentation
Phase 3: Donut chart, country chart and Top 5 customers

Phase 4: Matrix and Top 10 products chart
Phase 4: Category slicer and conditional formatting
Phase 5: Employee table, comparison chart and KPI cards
Phase 6: Sales trend, forecast and Smart Narrative
Phase 7: Four business KPI measures and regional revenue chart
Phase 8: Four report pages, slicers and text box
Save and test the PBIX file